

Lion Guardians Collared Report

Technical Guide

Collared Lion Movement Analysis — Methodology & Calculation Reference
Version 1.1 · Generated July 21, 2026

1. Overview

The **LG Collared Report** workflow analyses GPS-collared lion movement in the Amboseli ecosystem for the **Lion Guardians** programme. It ingests telemetry from **EarthRanger**, computes home range and track maps per subject, generates speed and distance statistics, and delivers an interactive dashboard plus a print-ready Word report.

Note: All per-group outputs (maps, metrics, Word sections) are produced by iterating over a user-chosen grouper: subject name, sex, or subtype.

2. Dependencies & Prerequisites

2.1 EarthRanger Connection

All telemetry is fetched from an **EarthRanger** instance via `set_er_connection`. The workflow calls `get_subjectgroup_observations` once over the configured time range. The filter: clean parameter discards junk-flagged observations before they reach the workflow.

2.2 Groupers

Three grouper fields are available via `set_groupers`:

Grouper field	EarthRanger source	Typical use
subject_name	subject.name	One output per individual collared lion — pre-selected by default
subject_sex	subject.sex	Aggregate by sex (M / F)
subject_subtype	subject.subject_subtype	Aggregate by subtype

2.3 Study Area Layers (from EarthRanger)

Unlike some other STE workflows, the boundary layers here are **not** downloaded from Dropbox — they are fetched live from the connected EarthRanger instance as spatial features via `get_spatial_features` (`select_geo_er`), then composited into a single static layer set by `create_spatial_features_layer` (`spatial_features_layer`). Only two feature types are queried:

Feature type	Style	Grouped by
Conservancies	Filled, sage green #8fbc8b, 75 % fill opacity	type_name
Group Ranch Boundaries	Unfilled, black outline, 1.25 px stroke width	type_name

Note: There is no separate conflict-hotspot layer in the current spec — every subject-level map is composited against exactly these two EarthRanger-sourced layers via `combine_deckgl_map_layers`.

2.4 Word Document Templates & Logo

File	Source	Purpose
collared_lions_cover_page.docx	Dropbox	Report cover page template — subject count, time range, preparer
collared_lion_subject_template.docx	Dropbox	Per-grouper section template — metrics and map images

File	Source	Purpose
lion-guardians.png	Dropbox	Fixed organisation logo used on the cover page — not user-configurable

All three files use `overwrite_existing: false` with 3 retries.

2.5 Base Map Tile Layers

Layer	Opacity	Max zoom
ArcGIS World Hillshade	100 %	15
ArcGIS World Street Map	15 %	15

The hillshade provides full-opacity terrain context. The street map is overlaid at 15 % to show road networks and settlement names without obscuring the hillshade or data layers.

3. Data Ingestion Pipeline

3.1 Observations → Relocations

`process_relocations` converts raw EarthRanger observations to a standardised GeoDataFrame. Retained columns:

Column	Source field	Description
<code>groupby_col</code>	<code>internal</code>	Subject identifier for grouping
<code>fixtime</code>	<code>observation timestamp</code>	UTC datetime of the GPS fix
<code>junk_status</code>	EarthRanger flag	True if fix is marked junk
<code>geometry</code>	<code>lat/lon</code>	Point geometry (WGS 84)
<code>extra__subject__name</code>	<code>subject.name</code>	Lion display name
<code>extra__subject__subject_subtype</code>	<code>subject.subject_subtype</code>	Subject subtype
<code>extra__subject__sex</code>	<code>subject.sex</code>	Subject sex (M / F / unknown)

Three bad coordinate pairs are removed unconditionally:

- (180.0, 90.0) — boundary sentinel
- (0.0, 0.0) — null-island artefact
- (1.0, 1.0) — common default / test value

Cleaned relocations are persisted as `relocations.geoparquet`.

3.2 Relocations → Trajectories

`relocations_to_trajectory` connects consecutive fixes per subject into `LineString` segments, adding `dist_meters`, `speed_kmh`, `segment_start`, and `segment_end`. A `trajectory_segment_filter` removes biologically implausible segments before analysis:

Filter parameter	Default	Description
min_length_meters	10	Discard segments shorter than 10 m
max_length_meters	10 000	Discard segments longer than 10 km
min_time_secs	10	Discard segments shorter than 10 s
max_time_secs	21 600	Discard segments longer than 6 hours
min_speed_kmh	1	Discard segments below 1 km/h average speed
max_speed_kmh	30	Discard segments above 30 km/h average speed

Trajectories are persisted as trajectories.geoparquet.

3.3 Temporal Index & Speed Classification

add_temporal_index keys the trajectory GeoDataFrame to segment_start, grouped by the configured groupers, enabling per-group iteration. apply_classification then bins speed_kmh into **6 equal-interval classes** (output column: speed_bins, labels to 1 decimal place).

3.4 Column Renaming & Group Splitting

Three columns are renamed via map_columns (raise_if_not_found: true):

Original column	Renamed to
extra__name	subject_name
extra__sex	subject_sex
extra__subject_subtype	subject_subtype

The renamed GeoDataFrame is split into per-group partitions by split_groups. All downstream map and metric tasks iterate over these partitions via mapvalues.

4. Static Map Layers

The two EarthRanger-sourced study area layers (Section 2.3) are built once and composited onto every subject-level map to provide spatial context.

4.1 Layer Styles

Layer	Colour	Opacity	Filled	Notes
Conservancies	#8fbc8b sage green	75 %	Yes	Grouped by type_name, legend title “Map Layers”
Group Ranch Boundaries	#000000 black outline	0 % fill	No	Outline only, stroke width 1.25

5. Map Outputs — Methodology

5.1 Home Range Map (Elliptical Time Density)

calculate_elliptical_time_density computes an ETD kernel density estimate per group, accounting for temporal autocorrelation in GPS telemetry. Parameters:

Parameter	Value	Meaning
CRS	ESRI:53042	World Azimuthal Equidistant — equal-area, minimises distortion
percentiles	50, 60, 70, 80, 90, 95, 99	Contour probability thresholds extracted as polygons
max_speed_factor	1.05	Estimated max speed as a multiple of the measured maximum; bounds the kernel spread
expansion_factor	1.3	Shape buffer expansion — controls how far density values spread across the grid
band_count	1	Single density band output
nodata_value	NaN	Cells outside probability mass are transparent

Contour polygons are coloured with the **RdYlGn** diverging colormap (innermost 50th percentile = red, outermost 99th = green). Map opacity is 45 %, legend title: *Home Range Percentiles*.

The map view is auto-zoomed via envelope_gdf (zoom_to_envelope) + compute_view_state_from_gdf (gdf_image_extent) — pitch 0, bearing 0, max zoom 15. An interactive HTML is persisted with a filename ending in _homerange.html, then converted to PNG via html_to_png (2× device scale factor, 40 s tile-load wait).

5.2 Subject Tracks Map

create_path_layer renders all trajectory segments per group as a continuous blue path layer:

Property	Value
Colour	RGB(30, 144, 255) — dodger blue
Width	1.25 px, min 2 px, max 8 px (screen-space)
Cap / Join	Rounded
Opacity	45 %

The path layer is combined with the two static study area layers via combine_deckgl_map_layers. An interactive HTML is persisted (suffix: tracks) and converted to PNG using the same screenshot settings as the Home Range map.

6. Summary Metrics

6.1 Per-Group Summary Table

summarize_df aggregates trajectory data grouped by subject_name:

Output column	Source	Aggregator	Unit	D.p.
mean_speed	speed_kmhr	mean	km/h	2
min_speed	speed_kmhr	min	km/h	2

Output column	Source	Aggregator	Unit	D.p.
max_speed	speed_kmph	max	km/h	2
total_distance	dist_meters	sum	m → km	2

The summary DataFrame (reset_index: true) is persisted directly to CSV — one row per subject, with no separate totals row appended.

6.2 Scalar Widget Values

Four values are derived per group via dataframe_column_sum then rounded to 2 d.p. by round_off_values:

- Mean Speed (km/h)
- Min Speed (km/h)
- Max Speed (km/h)
- Distance covered (km)

6.3 Unique Subject Count

dataframe_column_nunique counts distinct values in the subject_name column of the renamed trajectory GeoDataFrame (prior to the per-group split). This total is used in the cover page context.

7. Word Report (.docx)

7.1 Cover Page

prepare_cover_metadata (create_cover_tpl_context) builds the cover context (subject count, time range, *Ecoscope* as preparer, the fixed Lion Guardians logo). create_context_page (persist_collar_cover_page) renders it into cover_page.docx.

7.2 Per-Grouper Sections

ecoscope_workflows_ext_lion_guardians.tasks.reporting.create_collared_lions_context (build_collared_context) assembles a context dict per group containing the trajectory DataFrame, total distance, Home Range PNG, and Tracks PNG. ecoscope_workflows_ext_lion_guardians.tasks.reporting.render_docx_page (create_grouper_doc) renders each section from the subject template. Image boxes: **11.11 × 6.5 cm**. strict_images: true catches missing PNGs before rendering.

7.3 Document Merge

merge_docx_documents (merge_collared_docs) concatenates the cover page and all per-grouper sections into a single Word file, saved as overall_report.docx.

8. Interactive Dashboard

gather_dashboard assembles the **Lion Guardians** dashboard from six widget groups:

Widget	Type	Source
Mean Speed	Single value	round_mean_speed → total_mean_speed_sv_widget

Widget	Type	Source
Min Speed	Single value	round_min_speed → total_min_speed_sv_widget
Max Speed	Single value	round_max_speed → total_max_speed_sv_widget
Distance covered	Single value	round_total_distance → total_distance_sv_widget
Home Range Metrics	Map	td_ecomap_html_url → td_grouped_map_widget
Subject tracks	Map	track_html_url → sm_grouped_map_widget

Note: Widget tasks use skipif: [never] so the dashboard always assembles, even when some groups have no data.

9. Output Files

All files are written to \$ECOSCOPE_WORKFLOWS_RESULTS.

File / pattern	Format	Content
relocations.geoparquet	GeoParquet	Cleaned GPS fix locations
trajectories.geoparquet	GeoParquet	Segments with speed_kmhr, dist_meters, speed_bins
_homerange.html	HTML	Interactive ETD home range map
_tracks.html	HTML	Interactive subject tracks map
_homerange.png	PNG	2× screenshot of home range map
_tracks.png	PNG	2× screenshot of tracks map
_summary.csv	CSV	Speed and distance summary table (one row per subject)
cover_page.docx	Word	Rendered report cover page
.docx	Word	Per-grouper report section
overall_report.docx	Word	Final combined report

10. Workflow Execution Logic

10.1 Skip Conditions

Two default skip conditions apply to every task (task-instance-defaults):

- **any_is_empty_df** — skips the task (and all dependants) when any input DataFrame is empty, handling subjects with no fixes gracefully.
- **any_dependency_skipped** — propagates skips downstream automatically.

Widget and map-widget tasks override this with skipif: [never] to ensure the dashboard always assembles.

10.2 Data Flow Summary

Stage	Tasks
Setup	EarthRanger connection, time range, groupers, base maps

Stage	Tasks
Template download	2 Word templates + 1 logo PNG from Dropbox
Static layers	Conservancies + Group Ranch Boundaries fetched live from EarthRanger
Telemetry ingest	Observations → relocations → trajectories → speed bins → rename → split
Home Range branch	ETD → colormap → map layer → compose → view state → HTML → PNG → widget
Tracks branch	Path layer → compose → draw map → HTML → PNG → widget
Metrics branch	Summary table → totals row → CSV → scalar widget series
Report assembly	Cover page + per-group sections → merge docx
Dashboard	gather_dashboard combines all widgets

11. Software Versions

As of the migration to the **ecoscope-platform** task/compiler runtime, the workflow's requirements are:

Package	Version	Role
ecoscope-platform	>=2.15.0, <2.16.0	Core task library and workflow engine
ecoscope-workflows-ext-custom	0.1.0rc14.*	Utility tasks (column mapping, rounding, screenshots)
ecoscope-workflows-ext-ste	0.0.0rc1.*	Spatial-features and summary-table tasks
ecoscope-workflows-ext-lion-guardians	0.0.0rc1.*	Lion Guardians Word report rendering tasks
pydeck	0.9.2	Renders the interactive DeckGL maps
opentelemetry-sdk	>=1.20.0, <2.0.0	Workflow run telemetry/observability instrumentation

Note: ecoscope-workflows-core, ecoscope-workflows-ext-ecoscope, ecoscope-workflows-ext-mnc, and ecoscope-workflows-ext-big-life are not dependencies of this workflow.

Packages are distributed via the prefix.dev conda channel and pinned to compatible versions. The runtime environment is managed by **pixi**.